



Date: 04/06/2025

Lab Practical #01:

Study of basic networking commands and IP configuration.

Practical Assignment #01:

1. Perform and explain various networking commands listed below:
 - i. ipconfig
 - ii. ping
 - iii. getmac
 - iv. systeminfo
 - v. traceroute / tracert
 - vi. netstat
 - vii. nslookup
 - viii. hostname
 - ix. pathping
 - x. arp

1. ipconfig

Description:

Displays all current TCP/IP network configuration values and refreshes DHCP and DNS settings.

No.	Option	Description
1	/all	Displays full configuration information.
2	/release	Releases the IP address for the specified adapter.
3	/renew	Renews the IP address for the specified adapter.
4	/flushdns	Clears the DNS Resolver cache.
5	/displaydns	Displays the contents of the DNS Resolver cache.

Implementation:



Date: 04/06/2025

```
C:\Users\WELCOME>ipconfig /all
```

Windows IP Configuration

```
Host Name . . . . . : KALP-VIRADIA
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
```

Ethernet adapter Ethernet:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Realtek PCIe GbE Family Controller
Physical Address. . . . . : 04-BF-1B-5B-8B-24
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

Wireless LAN adapter Local Area Connection* 11:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #3
Physical Address. . . . . : 78-AF-08-18-AC-69
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

Wireless LAN adapter Local Area Connection* 12:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #4
Physical Address. . . . . : 7A-AF-08-18-AC-68
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

Wireless LAN adapter Wi-Fi 2:

```
Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) Wi-Fi 6 AX201 160MHz
Physical Address. . . . . : 78-AF-08-18-AC-68
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::2caf:7d11:3145:7963%10(Preferred)
IPv4 Address. . . . . : 10.20.66.235(Preferred)
Subnet Mask . . . . . : 255.255.0.0
Lease Obtained. . . . . : 04 June 2025 12:44:26 PM
Lease Expires . . . . . : 05 June 2025 12:44:26 PM
Default Gateway . . . . . : 10.20.1.1
DHCP Server . . . . . : 10.20.1.1
DHCPv6 IAID . . . . . : 360230664
DHCPv6 Client DUID. . . . . : 00-01-00-01-2F-BB-0C-48-04-BF-1B-5B-8B-24
DNS Servers . . . . . : 10.20.1.1
                        8.8.8.8
NetBIOS over Tcpi . . . . . : Enabled
```



Date: 04/06/2025

```
C:\Users\WELCOME>ipconfig /release
```

Windows IP Configuration

No operation can be performed on Ethernet while it has its media disconnected.
No operation can be performed on Local Area Connection* 11 while it has its media disconnected.

Ethernet adapter Ethernet:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 11:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 12:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi 2:

Connection-specific DNS Suffix . :
IPv6 Address. : 2409:40c1:3101:6088:aa69:93f3:b86:6c86
Temporary IPv6 Address. : 2409:40c1:3101:6088:c908:852f:eec6:4ee7
Link-local IPv6 Address : fe80::2caf:7d11:3145:7963%10
Default Gateway : fe80::a09c:eeff:feaf:7fe%10

```
C:\Users\WELCOME>ipconfig /renew
```

Windows IP Configuration

No operation can be performed on Ethernet while it has its media disconnected.
No operation can be performed on Local Area Connection* 11 while it has its media disconnected.
No operation can be performed on Local Area Connection* 12 while it has its media disconnected.

Ethernet adapter Ethernet:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 11:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 12:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi 2:

Connection-specific DNS Suffix . :
Link-local IPv6 Address : fe80::2caf:7d11:3145:7963%10
IPv4 Address. : 10.20.66.235
Subnet Mask : 255.255.0.0
Default Gateway : 10.20.1.1



Date: 04/06/2025

```
C:\Users\WELCOME>ipconfig /flushdns
```

```
Windows IP Configuration
```

```
Successfully flushed the DNS Resolver Cache.
```

```
C:\Users\WELCOME>ipconfig /displaydns
```

```
Windows IP Configuration
```

2. ping

Description:

Sends ICMP Echo Request messages to verify if a host is reachable over the network.

No.	Option	Description
1	-t	Pings the target until stopped manually.
2	-n count	Specifies the number of Echo Request messages to send.
3	-l size	Sends packets with a custom byte size.
4	-4	Forces using IPv4.
5	-6	Forces using IPv6.

Implementation:



Date: 04/06/2025

```
C:\Users\WELCOME>ping -t google.com
```

```
Pinging google.com [2404:6800:4009:823::200e] with 32 bytes of data:
```

```
Reply from 2404:6800:4009:823::200e: time=76ms
Reply from 2404:6800:4009:823::200e: time=87ms
Reply from 2404:6800:4009:823::200e: time=110ms
Reply from 2404:6800:4009:823::200e: time=93ms
Reply from 2404:6800:4009:823::200e: time=117ms
Reply from 2404:6800:4009:823::200e: time=108ms
Reply from 2404:6800:4009:823::200e: time=84ms
Reply from 2404:6800:4009:823::200e: time=105ms
Reply from 2404:6800:4009:823::200e: time=131ms
Reply from 2404:6800:4009:823::200e: time=106ms
Reply from 2404:6800:4009:823::200e: time=85ms
Reply from 2404:6800:4009:823::200e: time=106ms
Reply from 2404:6800:4009:823::200e: time=89ms
Reply from 2404:6800:4009:823::200e: time=109ms
Reply from 2404:6800:4009:823::200e: time=88ms
Reply from 2404:6800:4009:823::200e: time=77ms
Reply from 2404:6800:4009:823::200e: time=96ms
Reply from 2404:6800:4009:823::200e: time=82ms
```

```
Ping statistics for 2404:6800:4009:823::200e:
```

```
    Packets: Sent = 18, Received = 18, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 76ms, Maximum = 131ms, Average = 97ms
```

```
Control-C
```

```
^C
```

```
C:\Users\WELCOME>ping /n 4 google.com
```

```
Pinging google.com [142.250.70.110] with 32 bytes of data:
```

```
Reply from 142.250.70.110: bytes=32 time=23ms TTL=116
```

```
Reply from 142.250.70.110: bytes=32 time=25ms TTL=116
```

```
Reply from 142.250.70.110: bytes=32 time=22ms TTL=116
```

```
Reply from 142.250.70.110: bytes=32 time=22ms TTL=116
```

```
Ping statistics for 142.250.70.110:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 22ms, Maximum = 25ms, Average = 23ms
```



Date: 04/06/2025

```
C:\Users\WELCOME>ping /l 64 google.com

Pinging google.com [142.250.70.110] with 64 bytes of data:
Reply from 142.250.70.110: bytes=64 time=25ms TTL=116
Reply from 142.250.70.110: bytes=64 time=24ms TTL=116
Reply from 142.250.70.110: bytes=64 time=31ms TTL=116
Reply from 142.250.70.110: bytes=64 time=23ms TTL=116

Ping statistics for 142.250.70.110:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 23ms, Maximum = 31ms, Average = 25ms
```

```
C:\Users\WELCOME>ping -4 google.com

Pinging google.com [142.250.183.110] with 32 bytes of data:
Reply from 142.250.183.110: bytes=32 time=59ms TTL=112
Reply from 142.250.183.110: bytes=32 time=63ms TTL=112
Reply from 142.250.183.110: bytes=32 time=57ms TTL=112
Reply from 142.250.183.110: bytes=32 time=95ms TTL=112

Ping statistics for 142.250.183.110:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 57ms, Maximum = 95ms, Average = 68ms
```

```
C:\Users\WELCOME>ping -6 google.com

Pinging google.com [2404:6800:4009:823::200e] with 32 bytes of data:
Reply from 2404:6800:4009:823::200e: time=115ms
Reply from 2404:6800:4009:823::200e: time=81ms
Reply from 2404:6800:4009:823::200e: time=91ms
Reply from 2404:6800:4009:823::200e: time=116ms

Ping statistics for 2404:6800:4009:823::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 81ms, Maximum = 116ms, Average = 100ms
```

3. getmac

Description:

Displays the MAC address of the system's network interfaces.

No.	Option	Description
1	/v	Shows verbose output with more details.



Date: 04/06/2025

2	/fo table	Displays output in table format.
3	/fo list	Displays output in list format.
4	/fo csv	Displays output in CSV format.
5	/nh	Hides the column headers in the output.

Implementation:

```
C:\Users\WELCOME>getmac /v
```

```
Connection Name Network Adapter Physical Address Transport Name
=====
Ethernet Realtek PCIe Gb 04-BF-1B-5B-8B-24 Media disconnected
Wi-Fi 2 Intel(R) Wi-Fi 78-AF-08-18-AC-68 \Device\Tcpip_{7E118522-0AC2-4C9A-9566-F14BE7A7CB12}
```

```
C:\Users\WELCOME>getmac /fo table
```

```
Physical Address Transport Name
=====
04-BF-1B-5B-8B-24 Media disconnected
78-AF-08-18-AC-68 \Device\Tcpip_{7E118522-0AC2-4C9A-9566-F14BE7A7CB12}
```

```
C:\Users\WELCOME>getmac /fo list
```

```
Physical Address: 04-BF-1B-5B-8B-24
Transport Name: Media disconnected

Physical Address: 78-AF-08-18-AC-68
Transport Name: \Device\Tcpip_{7E118522-0AC2-4C9A-9566-F14BE7A7CB12}
```

```
C:\Users\WELCOME>getmac /fo csv
```

```
"Physical Address","Transport Name"
"04-BF-1B-5B-8B-24","Media disconnected"
"78-AF-08-18-AC-68","\Device\Tcpip_{7E118522-0AC2-4C9A-9566-F14BE7A7CB12}"
```

```
C:\Users\WELCOME>getmac /nh
```

```
04-BF-1B-5B-8B-24 Media disconnected
78-AF-08-18-AC-68 \Device\Tcpip_{7E118522-0AC2-4C9A-9566-F14BE7A7CB12}
```

4. systeminfo

Description:

Provides detailed configuration information about the computer and its OS.

No.	Option	Description
1	(no option)	Shows all system information.
2	/s	Specifies the remote system to connect to (hostname or IP address).



DARSHAN INSTITUTE OF ENGINEERING & TECHNOLOGY
Semester 5th | Practical Assignment | Computer Networks (2301CS501)

Date: 04/06/2025

3	/u	Specifies the user context to use for the connection.
4	/p	Specifies the password for the user account provided in /U. If omitted, it will prompt.
5	/fo	Displays the output in given format.

Implementation:

```
C:\Users\WELCOME>systeminfo

Host Name:                KALP-VIRADIA
OS Name:                  Microsoft Windows 11 Home Single Language
OS Version:               10.0.26100 N/A Build 26100
OS Manufacturer:         Microsoft Corporation
OS Configuration:        Standalone Workstation
OS Build Type:             Multiprocessor Free
Registered Owner:         WELCOME
Registered Organization:   N/A
Product ID:               00356-24680-32924-AAOEM
Original Install Date:     18-05-2025, 08:53:10 AM
System Boot Time:          04-06-2025, 12:44:04 PM
System Manufacturer:      Dell Inc.
System Model:              Dell G15 5520
System Type:               x64-based PC
Processor(s):              1 Processor(s) Installed.
                           [01]: Intel64 Family 6 Model 154 Stepping 3 GenuineIntel ~2500 Mhz
BIOS Version:              Dell Inc. 1.31.0, 05-02-2025
Windows Directory:         C:\WINDOWS
System Directory:          C:\WINDOWS\system32
Boot Device:                \Device\HarddiskVolume1
System Locale:              en-us;English (United States)
Input Locale:               00004009
Time Zone:                  (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Total Physical Memory:      24,261 MB
Available Physical Memory:  15,092 MB
Virtual Memory: Max Size:   25,797 MB
Virtual Memory: Available:  15,986 MB
Virtual Memory: In Use:     9,811 MB
Page File Location(s):      C:\pagefile.sys
Domain:                     WORKGROUP
Logon Server:               \\KALP-VIRADIA
```



DARSHAN INSTITUTE OF ENGINEERING & TECHNOLOGY

Semester 5th | Practical Assignment | Computer Networks (2301CS501)

Date: 04/06/2025

```
C:\Users\WELCOME>systeminfo /S KALP-VIRADIA
```

```
Host Name: KALP-VIRADIA
OS Name: Microsoft Windows 11 Home Single Language
OS Version: 10.0.26100 N/A Build 26100
OS Manufacturer: Microsoft Corporation
OS Configuration: Standalone Workstation
OS Build Type: Multiprocessor Free
Registered Owner: WELCOME
Registered Organization: N/A
Product ID: 00356-24680-32924-AAOEM
Original Install Date: 18-05-2025, 08:53:10 AM
System Boot Time: 02-07-2025, 02:17:51 PM
System Manufacturer: Dell Inc.
System Model: Dell G15 5520
System Type: x64-based PC
Processor(s): 1 Processor(s) Installed.
[01]: Intel64 Family 6 Model 154 Stepping 3 GenuineIntel ~2500 Mhz
BIOS Version: Dell Inc. 1.32.0, 26-03-2025
Windows Directory: C:\WINDOWS
System Directory: C:\WINDOWS\system32
Boot Device: \Device\HarddiskVolume1
System Locale: en-us;English (United States)
Input Locale: 00004009
Time Zone: (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Total Physical Memory: 24,261 MB
Available Physical Memory: 14,331 MB
Virtual Memory: Max Size: 25,797 MB
Virtual Memory: Available: 14,764 MB
Virtual Memory: In Use: 11,033 MB
Page File Location(s): C:\pagefile.sys
Domain: WORKGROUP
Logon Server: \\KALP-VIRADIA
Hotfix(s): 3 Hotfix(s) Installed.
[01]: KB5056579
[02]: KB5063060
[03]: KB5059502
Network Card(s): 3 NIC(s) Installed.
[01]: Realtek PCIe GbE Family Controller
Connection Name: Ethernet
```

```
C:\Users\WELCOME>systeminfo /S KALP-VIRADIA /U WELCOME
```

WARNING: User credentials cannot be used for local connections

```
Host Name: KALP-VIRADIA
OS Name: Microsoft Windows 11 Home Single Language
OS Version: 10.0.26100 N/A Build 26100
OS Manufacturer: Microsoft Corporation
OS Configuration: Standalone Workstation
OS Build Type: Multiprocessor Free
Registered Owner: WELCOME
Registered Organization: N/A
Product ID: 00356-24680-32924-AAOEM
Original Install Date: 18-05-2025, 08:53:10 AM
System Boot Time: 02-07-2025, 02:17:51 PM
System Manufacturer: Dell Inc.
System Model: Dell G15 5520
System Type: x64-based PC
Processor(s): 1 Processor(s) Installed.
[01]: Intel64 Family 6 Model 154 Stepping 3 GenuineIntel ~2500 Mhz
BIOS Version: Dell Inc. 1.32.0, 26-03-2025
Windows Directory: C:\WINDOWS
System Directory: C:\WINDOWS\system32
Boot Device: \Device\HarddiskVolume1
System Locale: en-us;English (United States)
Input Locale: 00004009
Time Zone: (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Total Physical Memory: 24,261 MB
Available Physical Memory: 14,358 MB
Virtual Memory: Max Size: 25,797 MB
Virtual Memory: Available: 14,769 MB
Virtual Memory: In Use: 11,028 MB
Page File Location(s): C:\pagefile.sys
Domain: WORKGROUP
Logon Server: \\KALP-VIRADIA
```



DARSHAN INSTITUTE OF ENGINEERING & TECHNOLOGY
Semester 5th | Practical Assignment | Computer Networks (2301CS501)

Date: 04/06/2025

```
C:\Users\WELCOME>systeminfo /S KALP-VIRADIA /U WELCOME /P 11412
WARNING: User credentials cannot be used for local connections

Host Name:                KALP-VIRADIA
OS Name:                   Microsoft Windows 11 Home Single Language
OS Version:               10.0.26100 N/A Build 26100
OS Manufacturer:         Microsoft Corporation
OS Configuration:        Standalone Workstation
OS Build Type:             Multiprocessor Free
Registered Owner:         WELCOME
Registered Organization:   N/A
Product ID:                00356-24680-32924-AAOEM
Original Install Date:     18-05-2025, 08:53:10 AM
System Boot Time:         02-07-2025, 02:17:51 PM
System Manufacturer:      Dell Inc.
System Model:              Dell G15 5520
System Type:               x64-based PC
Processor(s):              1 Processor(s) Installed.
                           [01]: Intel64 Family 6 Model 154 Stepping 3 GenuineIntel ~2500 Mhz
BIOS Version:              Dell Inc. 1.32.0, 26-03-2025
Windows Directory:        C:\WINDOWS
System Directory:          C:\WINDOWS\system32
Boot Device:               \Device\HarddiskVolume1
System Locale:              en-us;English (United States)
Input Locale:              00004009
Time Zone:                 (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Total Physical Memory:     24,261 MB
Available Physical Memory: 14,324 MB
Virtual Memory: Max Size: 25,797 MB
Virtual Memory: Available: 14,683 MB
Virtual Memory: In Use:    11,114 MB
Page File Location(s):     C:\pagefile.sys
Domain:                    WORKGROUP
Logon Server:              \\KALP-VIRADIA
Hotfix(s):                  3 Hotfix(s) Installed.
                           [01]: KB5056579
                           [02]: KB5063060
                           [03]: KB5059502
```



Date: 04/06/2025

```
C:\Users\WELCOME>systeminfo /S KALP-VIRADIA /U WELCOME /P 11412 /FO LIST
WARNING: User credentials cannot be used for local connections

Host Name:                        KALP-VIRADIA
OS Name:                          Microsoft Windows 11 Home Single Language
OS Version:                       10.0.26100 N/A Build 26100
OS Manufacturer:                 Microsoft Corporation
OS Configuration:                Standalone Workstation
OS Build Type:                    Multiprocessor Free
Registered Owner:                 WELCOME
Registered Organization:          N/A
Product ID:                       00356-24680-32924-AAOEM
Original Install Date:            18-05-2025, 08:53:10 AM
System Boot Time:                 02-07-2025, 02:17:51 PM
System Manufacturer:              Dell Inc.
System Model:                     Dell G15 5520
System Type:                       x64-based PC
Processor(s):                     1 Processor(s) Installed.
                                  [01]: Intel64 Family 6 Model 154 Stepping 3 GenuineIntel ~2500 Mhz
BIOS Version:                     Dell Inc. 1.32.0, 26-03-2025
Windows Directory:                C:\WINDOWS
System Directory:                 C:\WINDOWS\system32
Boot Device:                      \Device\HarddiskVolume1
System Locale:                     en-us;English (United States)
Input Locale:                     00004009
Time Zone:                        (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Total Physical Memory:             24,261 MB
Available Physical Memory:         14,387 MB
Virtual Memory: Max Size:         25,797 MB
Virtual Memory: Available:        14,801 MB
Virtual Memory: In Use:           10,996 MB
Page File Location(s):             C:\pagefile.sys
Domain:                           WORKGROUP
Logon Server:                     \\KALP-VIRADIA
Hotfix(s):                         3 Hotfix(s) Installed.
                                  [01]: KB5056579
                                  [02]: KB5063060
                                  [03]: KB5059502
```

5. traceroute / tracert

Description:

Traces the path that packets take to reach a network host.

No.	Option	Description
1	-d	Prevents resolving IPs to hostnames.
2	-h <max>	Sets the maximum number of hops.
3	-w <timeout>	Sets the timeout in milliseconds.
4	-4	Forces using IPv4.
5	-6	Forces using IPv6.

Implementation:



Date: 04/06/2025

```
C:\Users\WELCOME>tracert /d google.com
```

```
Tracing route to google.com [142.250.70.110]  
over a maximum of 30 hops:
```

1	2 ms	3 ms	2 ms	10.20.1.1
2	8 ms	5 ms	2 ms	180.211.109.177
3	6 ms	3 ms	4 ms	202.131.109.41
4	244 ms	813 ms	816 ms	120.72.95.129
5	6 ms	4 ms	3 ms	202.131.109.57
6	66 ms	19 ms	40 ms	202.131.99.106
7	22 ms	16 ms	16 ms	72.14.204.217
8	18 ms	18 ms	17 ms	142.251.76.27
9	23 ms	20 ms	25 ms	192.178.86.241
10	22 ms	19 ms	19 ms	142.250.70.110

```
Trace complete.
```

```
C:\Users\WELCOME>tracert /h 10 google.com
```

```
Tracing route to google.com [142.250.70.110]  
over a maximum of 10 hops:
```

1	2 ms	1 ms	2 ms	10.20.1.1
2	3 ms	2 ms	2 ms	180.211.109.177
3	*	24 ms	9 ms	202.131.109.41
4	9 ms	3 ms	3 ms	120.72.95.129
5	6 ms	5 ms	7 ms	202.131.109.57
6	19 ms	16 ms	16 ms	202.131.99.106
7	48 ms	87 ms	35 ms	72.14.204.217
8	51 ms	24 ms	27 ms	142.251.76.27
9	24 ms	21 ms	19 ms	192.178.86.241
10	23 ms	20 ms	30 ms	pnbomb-ac-in-f14.1e100.net [142.250.70.110]

```
Trace complete.
```



Date: 04/06/2025

```
C:\Users\WELCOME>tracert -w 2 google.com
```

```
Tracing route to google.com [2404:6800:4009:80b::200e]  
over a maximum of 30 hops:
```

1	4 ms	2 ms	3 ms	2409:40c1:3101:6088::41
2	42 ms	36 ms	37 ms	2405:200:5210:5:3924:110:3:307
3	*	*	*	Request timed out.
4	*	*	*	Request timed out.
5	*	*	*	Request timed out.
6	45 ms	35 ms	40 ms	2405:200:801:2e00::84
7	*	*	*	Request timed out.
8	*	*	*	Request timed out.
9	124 ms	121 ms	105 ms	2404:6800:812f::1
10	104 ms	*	69 ms	2404:6800:812f::1
11	120 ms	74 ms	78 ms	2404:6800:812f::1
12	128 ms	85 ms	*	2001:4860:0:1::539c
13	67 ms	74 ms	*	2001:4860:0:1::77d0
14	80 ms	89 ms	83 ms	2001:4860::9:4000:d773
15	74 ms	79 ms	*	2001:4860::c:4004:53cb
16	92 ms	115 ms	*	2001:4860::9:4002:d931
17	*	*	85 ms	2001:4860:0:1::1baf
18	77 ms	90 ms	72 ms	2001:4860:0:1::269f
19	*	80 ms	74 ms	pnbomb-ba-in-x0e.1e100.net [2404:6800:4009:80b::200e]

Trace complete.

```
C:\Users\WELCOME>tracert -4 google.com
```

```
Tracing route to google.com [142.251.220.46]  
over a maximum of 30 hops:
```

1	3 ms	9 ms	3 ms	192.168.102.69
2	43 ms	6 ms	5 ms	192.0.0.1
3	*	*	*	Request timed out.
4	*	*	*	Request timed out.
5	*	*	*	Request timed out.
6	*	*	*	Request timed out.
7	*	*	*	Request timed out.
8	*	*	*	Request timed out.
9	*	*	*	Request timed out.
10	*	*	*	Request timed out.
11	*	*	*	Request timed out.
12	*	*	*	Request timed out.
13	*	*	*	Request timed out.
14	80 ms	54 ms	38 ms	pnbomb-ba-in-f14.1e100.net [142.251.220.46]

Trace complete.



Date: 04/06/2025

```
C:\Users\WELCOME>tracert -6 google.com

Tracing route to google.com [2404:6800:4009:823::200e]
over a maximum of 30 hops:

  1    16 ms     9 ms     15 ms  2409:40c1:3101:6088::41
  2    47 ms    25 ms    28 ms  2405:200:5210:5:3924:110:3:307
  3      *       *       *    Request timed out.
  4      *       *       *    Request timed out.
  5      *       *       *    Request timed out.
  6    71 ms    37 ms    28 ms  2405:200:801:2e00::84
  7      *       *       *    Request timed out.
  8      *       *       *    Request timed out.
  9   122 ms    67 ms    68 ms  2405:200:802:760::8
 10   115 ms    94 ms    78 ms  2405:200:802:760::8
 11      *       *       *    Request timed out.
 12   104 ms    73 ms    77 ms  2404:6800:8281:240::1
 13   131 ms    71 ms    77 ms  2404:6800:8281:240::1
 14      *       *       *    Request timed out.
 15      *    116 ms     *    2001:4860:0:1::7976
 16    79 ms    78 ms    78 ms  2001:4860:0:1::8767
 17   108 ms    72 ms    78 ms  2001:4860:0:1::f69
 18   102 ms    78 ms    85 ms  bom12s13-in-x0e.1e100.net [2404:6800:4009:823::200e]

Trace complete.
```

6. netstat

Description:

Displays network connections, routing tables, interface stats, etc.

No.	Option	Description
1	-a	Displays all connections and listening ports.
2	-n	Displays addresses and port numbers numerically.
3	-o	Displays owning process ID associated with each connection.
4	-e	Displays Ethernet statistics.
5	-s	Displays per-protocol statistics.

Implementation:



Date: 04/06/2025

```
C:\Windows\System32>netstat -a
```

Active Connections

Proto	Local Address	Foreign Address	State
TCP	0.0.0.0:135	KALP-VIRADIA:0	LISTENING
TCP	0.0.0.0:445	KALP-VIRADIA:0	LISTENING
TCP	0.0.0.0:5040	KALP-VIRADIA:0	LISTENING
TCP	0.0.0.0:7070	KALP-VIRADIA:0	LISTENING
TCP	0.0.0.0:49664	KALP-VIRADIA:0	LISTENING
TCP	0.0.0.0:49665	KALP-VIRADIA:0	LISTENING
TCP	0.0.0.0:49668	KALP-VIRADIA:0	LISTENING
TCP	0.0.0.0:49669	KALP-VIRADIA:0	LISTENING
TCP	0.0.0.0:49672	KALP-VIRADIA:0	LISTENING
TCP	0.0.0.0:49691	KALP-VIRADIA:0	LISTENING
TCP	10.20.68.170:139	KALP-VIRADIA:0	LISTENING
TCP	10.20.68.170:49408	4.213.25.240:https	ESTABLISHED
TCP	10.20.68.170:49722	relay-0645e242:http	ESTABLISHED
TCP	10.20.68.170:49836	13.107.246.68:https	CLOSE_WAIT
TCP	10.20.68.170:49884	40.104.66.98:https	ESTABLISHED
TCP	10.20.68.170:49885	40.104.66.98:https	ESTABLISHED



DARSHAN INSTITUTE OF ENGINEERING & TECHNOLOGY
Semester 5th | Practical Assignment | Computer Networks (2301CS501)

Date: 04/06/2025

```
C:\Users\WELCOME>netstat -n
```

Active Connections

Proto	Local Address	Foreign Address	State
TCP	127.0.0.1:49724	127.0.0.1:49725	ESTABLISHED
TCP	127.0.0.1:49725	127.0.0.1:49724	ESTABLISHED
TCP	127.0.0.1:49726	127.0.0.1:49727	ESTABLISHED
TCP	127.0.0.1:49727	127.0.0.1:49726	ESTABLISHED
TCP	127.0.0.1:49728	127.0.0.1:49729	ESTABLISHED
TCP	127.0.0.1:49729	127.0.0.1:49728	ESTABLISHED
TCP	127.0.0.1:49732	127.0.0.1:49733	ESTABLISHED
TCP	127.0.0.1:49733	127.0.0.1:49732	ESTABLISHED
TCP	127.0.0.1:49859	127.0.0.1:49860	ESTABLISHED
TCP	127.0.0.1:49860	127.0.0.1:49859	ESTABLISHED
TCP	192.168.102.91:50124	148.113.16.130:80	ESTABLISHED
TCP	[2409:40c1:3101:6088:c908:852f:eec6:4ee7]:49414	[2603:1040:a06:6::2]:443	ESTABLISHED
TCP	[2409:40c1:3101:6088:c908:852f:eec6:4ee7]:49415	[2603:1040:a06:6::1]:443	ESTABLISHED
TCP	[2409:40c1:3101:6088:c908:852f:eec6:4ee7]:50126	[2404:6800:4003:c06:bc]:5228	ESTABLISHED
TCP	[2409:40c1:3101:6088:c908:852f:eec6:4ee7]:50173	[2606:4700:8ca2:4094:cc0c:ae6:f9fb:48a1]:443	TIME_WAIT
TCP	[2409:40c1:3101:6088:c908:852f:eec6:4ee7]:50187	[2404:6800:4009:830::2001]:443	ESTABLISHED

```
C:\Users\WELCOME>netstat -o
```

Active Connections

Proto	Local Address	Foreign Address	State	PID
TCP	127.0.0.1:49724	KALP-VIRADIA:49725	ESTABLISHED	1720
TCP	127.0.0.1:49725	KALP-VIRADIA:49724	ESTABLISHED	1720
TCP	127.0.0.1:49726	KALP-VIRADIA:49727	ESTABLISHED	1912
TCP	127.0.0.1:49727	KALP-VIRADIA:49726	ESTABLISHED	1912
TCP	127.0.0.1:49728	KALP-VIRADIA:49729	ESTABLISHED	3532
TCP	127.0.0.1:49729	KALP-VIRADIA:49728	ESTABLISHED	3532
TCP	127.0.0.1:49732	KALP-VIRADIA:49733	ESTABLISHED	5488
TCP	127.0.0.1:49733	KALP-VIRADIA:49732	ESTABLISHED	5488
TCP	127.0.0.1:49859	KALP-VIRADIA:49860	ESTABLISHED	11916
TCP	127.0.0.1:49860	KALP-VIRADIA:49859	ESTABLISHED	11916
TCP	192.168.102.91:50124	relay-8458821c:http	ESTABLISHED	5472
TCP	[2409:40c1:3101:6088:c908:852f:eec6:4ee7]:49414	[2603:1040:a06:6::2]:https	ESTABLISHED	5552
TCP	[2409:40c1:3101:6088:c908:852f:eec6:4ee7]:49415	[2603:1040:a06:6::1]:https	ESTABLISHED	5552
TCP	[2409:40c1:3101:6088:c908:852f:eec6:4ee7]:50126	sm-in-f188:5228	ESTABLISHED	21744
TCP	[2409:40c1:3101:6088:c908:852f:eec6:4ee7]:50173	[2606:4700:8ca2:4094:cc0c:ae6:f9fb:48a1]:https	TIME_WAIT	0
TCP	[2409:40c1:3101:6088:c908:852f:eec6:4ee7]:50187	bom12s20-in-x01:https	ESTABLISHED	14364

```
C:\Users\WELCOME>netstat -e
```

Interface Statistics

	Received	Sent
Bytes	339407346	14091510
Unicast packets	283050	61350
Non-unicast packets	6	2118
Discards	0	0
Errors	0	0
Unknown protocols	0	



Date: 04/06/2025

```
C:\Users\WELCOME>netstat -s
```

IPv4 Statistics

Packets Received	= 2566
Received Header Errors	= 0
Received Address Errors	= 0
Datagrams Forwarded	= 0
Unknown Protocols Received	= 0
Received Packets Discarded	= 490
Received Packets Delivered	= 3269
Output Requests	= 4006
Routing Discards	= 0
Discarded Output Packets	= 171
Output Packet No Route	= 10
Reassembly Required	= 0
Reassembly Successful	= 0
Reassembly Failures	= 0
Datagrams Successfully Fragmented	= 0
Datagrams Failing Fragmentation	= 0
Fragments Created	= 0

IPv6 Statistics

Packets Received	= 34125
Received Header Errors	= 0
Received Address Errors	= 0
Datagrams Forwarded	= 0
Unknown Protocols Received	= 5
Received Packets Discarded	= 53
Received Packets Delivered	= 34199
Output Requests	= 28721
Routing Discards	= 0
Discarded Output Packets	= 37
Output Packet No Route	= 4
Reassembly Required	= 0
Reassembly Successful	= 0
Reassembly Failures	= 0
Datagrams Successfully Fragmented	= 0

7. nslookup

Description:

Queries the DNS to obtain domain name or IP address mapping.



Date: 04/06/2025

No.	Option	Description
1	<domain>	Retrieves IP address for the domain.
2	<IP>	Retrieves domain name for the given IP (reverse lookup).
3	set type=MX	Retrieves Mail Exchange records.
4	set type=NS	Retrieves Name Server records.
5	server <IP>	Specifies DNS server for the query.

Implementation:

```
C:\Users\WELCOME>nslookup google.com 1.1.1.1
Server:  one.one.one.one
Address:  1.1.1.1

Non-authoritative answer:
Name:     google.com
Addresses: 2404:6800:4009:80b::200e
          142.251.42.14
```

```
C:\Users\WELCOME>nslookup 8.8.8.8
Server:  UnKnown
Address: 192.168.102.69

Name:     dns.google
Address:  8.8.8.8
```



Date: 04/06/2025

```
C:\Users\WELCOME>nslookup -type=mx google.com
Server: UnKnown
Address: 192.168.102.69

Non-authoritative answer:
google.com      MX preference = 10, mail exchanger = smtp.google.com
```

```
C:\Users\WELCOME>nslookup -type=ns google.com
Server: UnKnown
Address: 192.168.102.69

Non-authoritative answer:
google.com      nameserver = ns2.google.com
google.com      nameserver = ns3.google.com
google.com      nameserver = ns1.google.com
google.com      nameserver = ns4.google.com

ns1.google.com  internet address = 216.239.32.10
ns1.google.com  AAAA IPv6 address = 2001:4860:4802:32::a
ns4.google.com  internet address = 216.239.38.10
ns4.google.com  AAAA IPv6 address = 2001:4860:4802:38::a
ns2.google.com  internet address = 216.239.34.10
ns2.google.com  AAAA IPv6 address = 2001:4860:4802:34::a
ns3.google.com  internet address = 216.239.36.10
ns3.google.com  AAAA IPv6 address = 2001:4860:4802:36::a
```

```
C:\Users\WELCOME>nslookup
Default Server: UnKnown
Address: 192.168.102.69

> server 8.8.8.8
Default Server: dns.google
Address: 8.8.8.8
```

8. hostname

Description:

Displays the current name of the computer.

No.	Option	Description
1	(no option)	Prints current hostname.
2		



Date: 04/06/2025

3		
4		
5		

Implementation:

```
C:\Users\WELCOME>hostname  
KALP-VIRADIA
```

9. pathping

Description:

Combines the functions of ping and tracert to provide information on packet loss at each hop.

No.	Option	Description
1	-n	Prevents resolving hostnames.
2	-h <max>	Sets maximum number of hops.
3	-g <host-list>	Specifies loose source route along host list.
4	-p <ms>	Wait time between pings.
5	-q <num>	Number of queries per hop.

Implementation:

```
C:\Users\WELCOME>pathping -n google.com  
  
Tracing route to google.com [2404:6800:4009:829::200e]  
over a maximum of 30 hops:  
 0  2409:40c1:3101:6088:c908:852f:eec6:4ee7  
 1  2409:40c1:3101:6088::41  
 2  2405:200:5210:5:3924:110:3:307  
 3      *           *           *  
Computing statistics for 50 seconds...  
Hop  RTT      Source to Here   This Node/Link   Address  
    Lost/Sent = Pct  Lost/Sent = Pct  
 0           0/ 100 = 0%    0/ 100 = 0%    2409:40c1:3101:6088:c908:852f:eec6:4ee7  
 1    4ms      0/ 100 = 0%    0/ 100 = 0%    2409:40c1:3101:6088::41  
 2   37ms      0/ 100 = 0%    0/ 100 = 0%    2405:200:5210:5:3924:110:3:307  
  
Trace complete.
```



Date: 04/06/2025

```
C:\Users\WELCOME>pathping -h 10 google.com
```

```
Tracing route to google.com [2404:6800:4009:829::200e]
```

```
over a maximum of 10 hops:
```

```
0 KALP-VIRADIA [2409:40c1:3101:6088:c908:852f:eec6:4ee7]
```

```
1 2409:40c1:3101:6088::41
```

```
2 2405:200:5210:5:3924:110:3:307
```

```
3 * * *
```

```
Computing statistics for 50 seconds...
```

Hop	RTT	Source to Here Lost/Sent = Pct	This Node/Link Lost/Sent = Pct	Address
0				KALP-VIRADIA [2409:40c1:3101:6088:c908:852f:eec6:4ee7]
1	7ms	0/ 100 = 0%	0/ 100 = 0%	2409:40c1:3101:6088::41
2	41ms	0/ 100 = 0%	0/ 100 = 0%	2405:200:5210:5:3924:110:3:307

```
Trace complete.
```

```
C:\Users\WELCOME>pathping -g 192.168.1.1 8.8.8.8 google.com
```

```
Tracing route to google.com [142.251.220.46]
```

```
over a maximum of 30 hops:
```

```
0 KALP-VIRADIA [192.168.102.91]
```

```
1 * * *
```

```
Computing statistics for 0 seconds...
```

Hop	RTT	Source to Here Lost/Sent = Pct	This Node/Link Lost/Sent = Pct	Address
0				KALP-VIRADIA [192.168.102.91]

```
Trace complete.
```



Date: 04/06/2025

```
C:\Users\WELCOME>pathping -p 500 google.com

Tracing route to google.com [2404:6800:4009:80b::200e]
over a maximum of 30 hops:
  0 KALP-VIRADIA [2409:40c1:3101:6088:c908:852f:eec6:4ee7]
  1 2409:40c1:3101:6088::41
  2 2405:200:5210:5:3924:110:3:307
  3 * * *
Computing statistics for 100 seconds...
Hop  RTT      Source to Here   This Node/Link   Address
    0      0/ 100 = 0%    0/ 100 = 0%    KALP-VIRADIA [2409:40c1:3101:6088:c908:852f:eec6:4ee7]
    1  25ms      0/ 100 = 0%    0/ 100 = 0%    2409:40c1:3101:6088::41
    2  35ms      0/ 100 = 0%    0/ 100 = 0%    2405:200:5210:5:3924:110:3:307

Trace complete.

C:\Users\WELCOME>pathping -q 3 google.com

Tracing route to google.com [2404:6800:4009:80b::200e]
over a maximum of 30 hops:
  0 KALP-VIRADIA [2409:40c1:3101:6088:c908:852f:eec6:4ee7]
  1 2409:40c1:3101:6088::41
  2 2405:200:5210:5:3924:110:3:307
  3 * * *
Computing statistics for 1 seconds...
Hop  RTT      Source to Here   This Node/Link   Address
    0      0/ 3 = 0%    0/ 3 = 0%    KALP-VIRADIA [2409:40c1:3101:6088:c908:852f:eec6:4ee7]
    1  20ms      0/ 3 = 0%    0/ 3 = 0%    2409:40c1:3101:6088::41
    2  69ms      0/ 3 = 0%    0/ 3 = 0%    2405:200:5210:5:3924:110:3:307

Trace complete.
```

10.arp

Description:

Displays and modifies the ARP (Address Resolution Protocol) cache.

No.	Option	Description
1	-a	Displays current ARP entries.
2	-g	Displays current ARP entries (same as -a).
3	-d <IP>	Deletes ARP entry for the specified IP address.
4	-s <IP> <MAC>	Adds a static ARP entry.
5	-v	Shows verbose output.

Implementation:



Date: 04/06/2025

```
C:\Users\WELCOME>arp -a
```

```
Interface: 192.168.102.91 --- 0xa
```

Internet Address	Physical Address	Type
192.168.102.69	a2-9c-ee-af-07-fe	dynamic
192.168.102.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

```
C:\Users\WELCOME>arp -g
```

```
Interface: 192.168.102.91 --- 0xa
```

Internet Address	Physical Address	Type
192.168.102.69	a2-9c-ee-af-07-fe	dynamic
192.168.102.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

```
C:\Windows\System32>arp -d 192.168.102.69
```

```
C:\Windows\System32>arp -s 192.168.1.100 00-aa-bb-cc-dd-ee
```



Date: 04/06/2025

```
C:\Windows\System32>arp -v
```

Displays and modifies the IP-to-Physical address translation tables used by address resolution protocol (ARP).

```
ARP -s inet_addr eth_addr [if_addr]
```

```
ARP -d inet_addr [if_addr]
```

```
ARP -a [inet_addr] [-N if_addr] [-v]
```

-a Displays current ARP entries by interrogating the current protocol data. If `inet_addr` is specified, the IP and Physical addresses for only the specified computer are displayed. If more than one network interface uses ARP, entries for each ARP table are displayed.

-g Same as **-a**.

-v Displays current ARP entries in verbose mode. All invalid entries and entries on the loop-back interface will be shown.

inet_addr Specifies an internet address.

-N if_addr Displays the ARP entries for the network interface specified by `if_addr`.

-d Deletes the host specified by `inet_addr`. `inet_addr` may be wildcarded with `*` to delete all hosts.

-s Adds the host and associates the Internet address `inet_addr` with the Physical address `eth_addr`. The Physical address is given as 6 hexadecimal bytes separated by hyphens. The entry is permanent.

eth_addr Specifies a physical address.

if_addr If present, this specifies the Internet address of the interface whose address translation table should be modified. If not present, the first applicable interface will be used.

Example:

```
> arp -s 157.55.85.212 00-aa-00-62-c6-09 .... Adds a static entry.
```

```
> arp -a .... Displays the arp table.
```